

ATLAS Rev. 1.4

Operating Manual

REV.1.0 17.01.2026 (English version)

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Introduction

The purpose of this User Manual is to provide clear and structured guidance on the correct use of the ATLAS Rev. 1.4 model, covering everything from basic start-up and operating procedures to more advanced aspects such as configuration, programming, and general recommendations for use. Reading and understanding this manual will not only ensure the safe operation of the robot, but also improve its performance and extend the useful life of its components.

This document is intended for students, enthusiasts, and competitors in robotics. It is recommended to follow the instructions in the order presented and pay special attention to the sections on warnings and important notes, as incorrect use can lead to malfunctions or damage to the equipment.

This model is constantly evolving; therefore, some procedures or parameters may vary depending on the hardware or firmware version. This manual will serve as the primary reference for understanding the robot's design philosophy and establishing a solid foundation for future modifications, improvements, or customizations. Translated with DeepL.com (free version)

General description – ATLAS

ATLAS is a series of high-performance line-following robots designed for advanced applications in research, development, and experimentation. Its architecture prioritizes functional simplicity, ease of use, and maintenance, integrating a set of features that make it a versatile platform for both studying and learning about this type of technology.

Thanks to a relatively simple but highly effective architecture based on the ATmega328P microcontroller, ATLAS provides a clear understanding of its internal workings, facilitating the learning of the fundamental principles that position it among the fastest line-following robots in the world.

The ATLAS model has been designed as a technical tool aimed at users with intermediate knowledge of programming, electronics, and robotics. Consequently, results beyond the limits and conditions specified in this user manual are not guaranteed, and the operator is responsible for its correct configuration, operation, and use.

General description – Composition

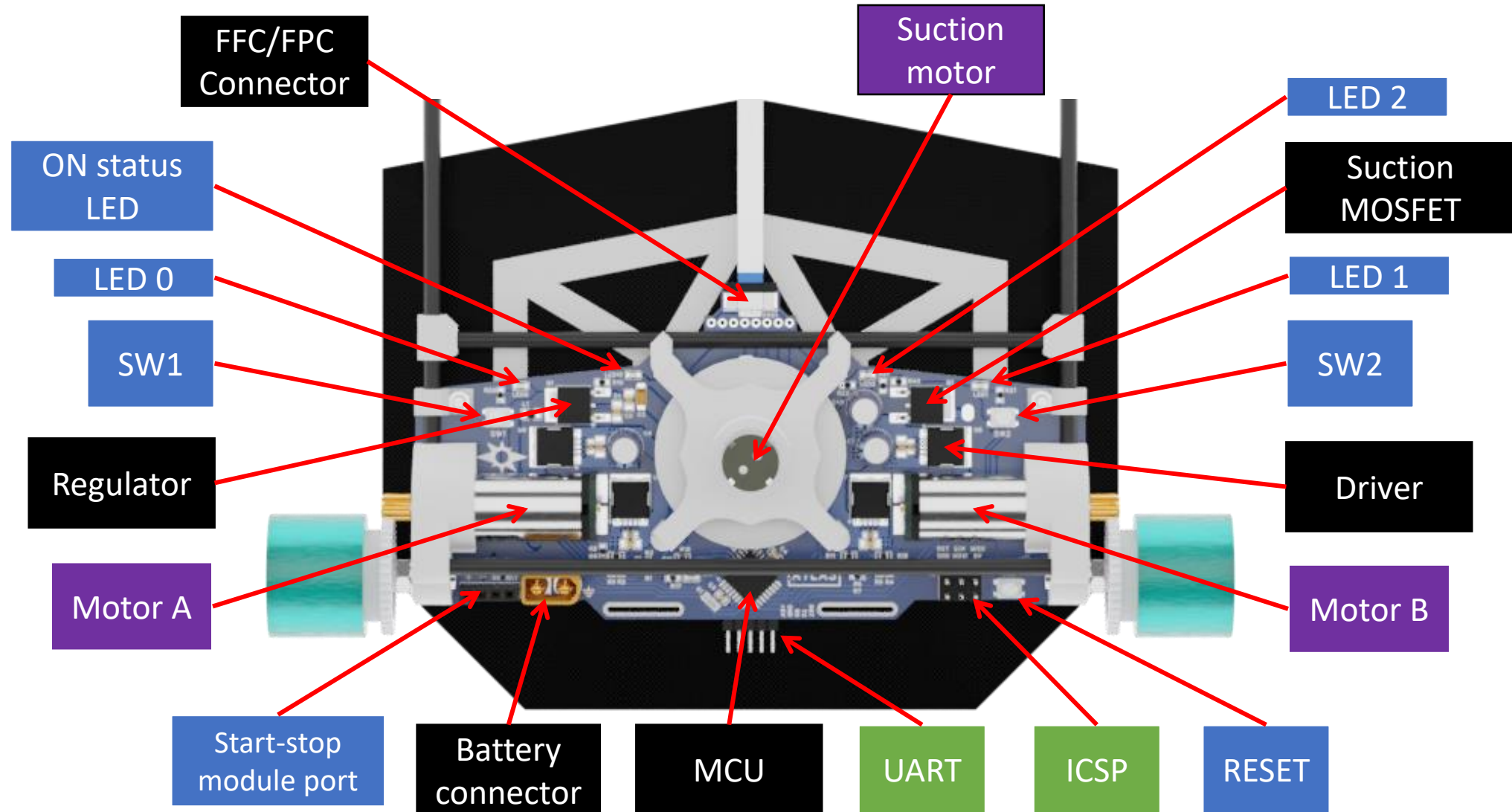
The ATLAS Rev. 1.4 model consists of a total of three 7.4 V DC 1023 motors: two for the differential drive system and one additional motor for the suction system. The drive motors are powered directly from the main supply voltage (V_s) via the XT30 connector, supporting an operating range of 7.4 V to 12.4 V (2S–3S). The suction system motor has a separate power connector, designed to operate at 7.4 V (2S), which must be connected directly to the battery charging port.

For user interaction, ATLAS incorporates only three physical buttons: SW1, SW2, and RESET, in addition to a dedicated terminal for connecting an external boot module. The visual signaling system consists of three programmable LEDs (LED0–LED2), which indicate the robot's operating or configuration status.

The robot has an ICSP port for direct microcontroller programming, as well as a UART terminal for serial communication. Additionally, it is possible to program the microcontroller through the UART interface, depending on the USB-serial converter used.

In order to facilitate use and prevent incorrect connections, all user-accessible peripherals are labeled on the PCB, indicating the polarity or name of the pins. The following page shows the general layout of the system peripherals.

General description – Peripherals distribution



Technical specifications

<i>Model:</i>	ATLAS Rev. 1.4
<i>Dimensions:</i>	17 x 24.5 x 4.5 cm
<i>Weight (without batt):</i>	93g
<i>Operation voltage:</i>	2-3 S
<i>Voltage regulator:</i>	78M05
<i>Motor drivers:</i>	BTN9960LV
<i>MCU:</i>	ATmega328P-AU
<i>Motors:</i>	3x coreless 1023 7.4V
<i>Line sensors:</i>	16x QRE1113GR

Software and drivers – Arduino IDE

The Arduino IDE version 2.3.6 development environment is used for programming and flashing the ATLAS series. This software can be downloaded directly from the official Arduino website via the following link: <https://www.arduino.cc/en/software/#ide>

Alternatively, you can access the official Arduino website and download the latest version of the IDE, which, in most cases, is compatible with the ATLAS series.

The installation process consists solely of running the downloaded file, accepting the license agreement, selecting the user and destination path, and continuing with the guided installation. The Arduino IDE requires approximately 550 MB of available hard disk space.

Once the installation is complete and the program is opened for the first time, the Arduino IDE itself will request the installation of the necessary dependencies and additional components. These notifications are displayed in the lower right section of the interface and must be addressed to ensure the proper functioning of the development environment and communication with the ATLAS series.



Software and drivers – Zadig

Zadig is an application designed to install generic USB drivers on Windows systems. In the case, it is specifically used to install the WinUSB driver, which is necessary for the USBasp programmer to function correctly during the microcontroller flashing process.

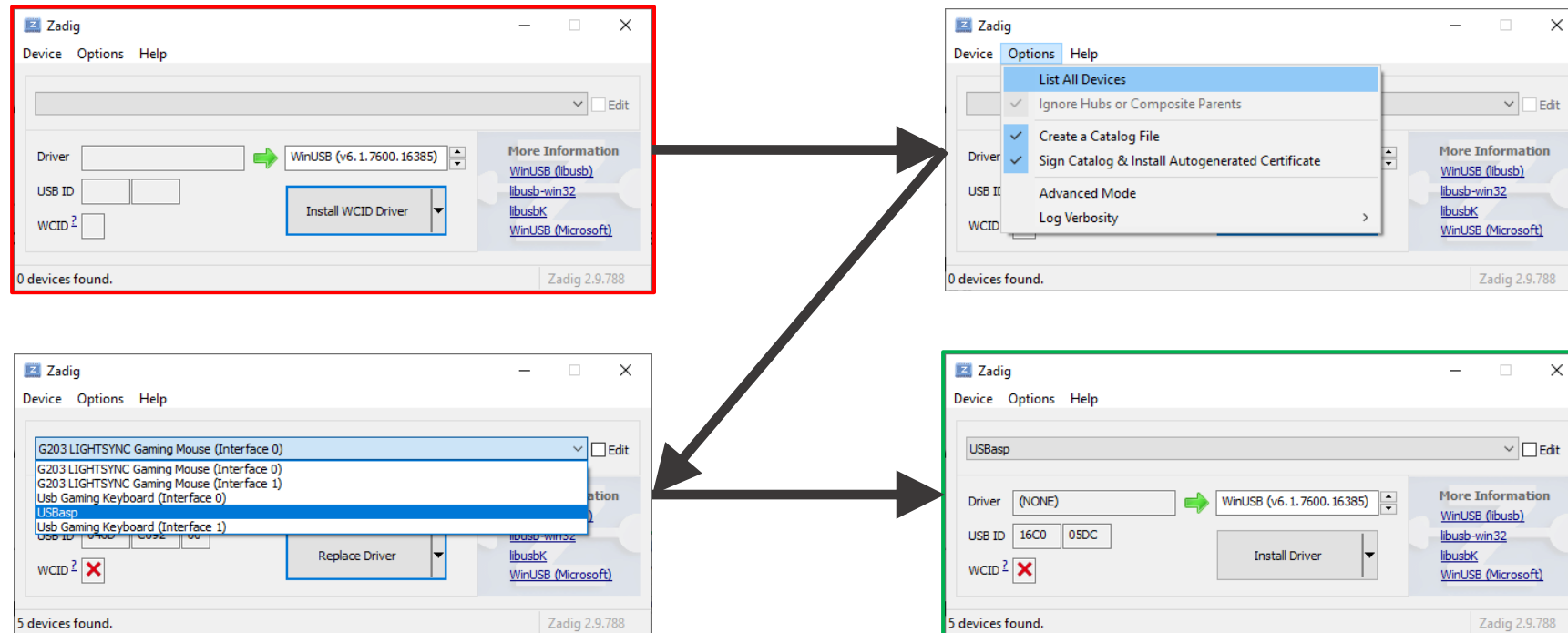
The steps required to install the corresponding driver are described below:

1. Download the application from the official website: <https://zadig.akeo.ie/>
2. Alternatively, you can search for it directly from your browser. At the time of writing this manual, the latest version is v2.9, with an approximate size of 5.1 MB.
3. Before running the program, make sure that the USBasp programmer is properly connected to your computer.
4. Run the downloaded file. The system will display a User Account Control (UAC) window requesting permission for the application to make changes to the device. Select the “Yes” option to continue.



Software and drivers – Zadig

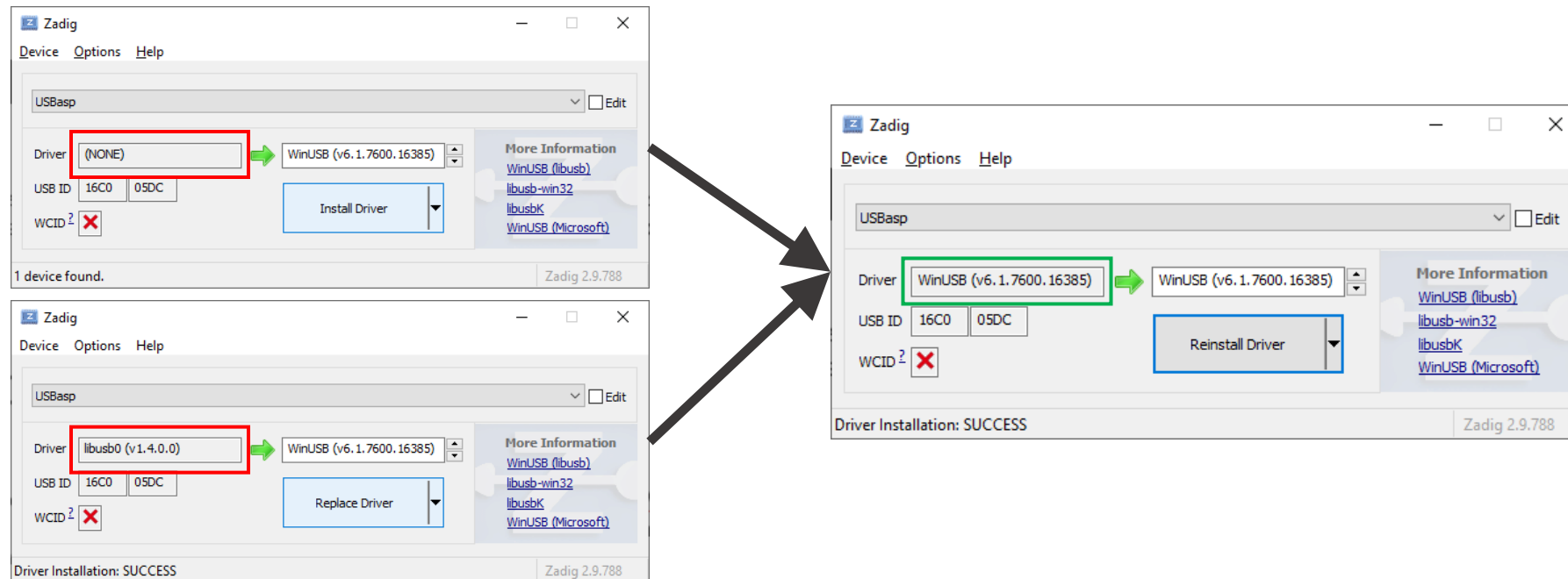
4. Once Zadig is open, the “USBasp” device should appear in the list of available devices. If it does not appear immediately, enable the “List All Devices” option. Verify that the USBasp is properly connected, then find and select “USBasp” from the drop-down list.



Software and drivers – Zadig

5. After selecting the correct device, the recommended driver “WinUSB (v6.1.7600.16385)” should automatically appear in the ‘Driver’ field. Press the “Install Driver” button to start the installation, which will take approximately 1 minute. If a different driver is already installed, select the WinUSB driver and press the “Replace Driver” option.

Once this process is complete, the USBasp programmer will be correctly configured for use with the ATLAS Rev. 1.x.x series.



Robot flashing – Requirements

The ATLAS_1.4.1.ino firmware incorporates an integrated interface that allows the line color to be selected prior to the calibration process, as well as adjusting the robot's operating speed. Additionally, it has an alternate debug mode (debugMode), which allows individual tests to be performed on each peripheral in order to verify its correct operation and facilitate the diagnosis of possible faults.

These features were developed with the aim of minimizing the need to reprogram the robot, as well as providing an intuitive operating menu for its configuration and activation. Therefore, if the default parameters meet the user's experimentation requirements, it is not necessary to flash the robot.

Before proceeding with the explanation of the programming method, it is essential to analyze and understand the general structure of the code in order to avoid unexpected behavior during operation or when modifying program parameters and sequences.

To flash the robot, the following requirements must be met.

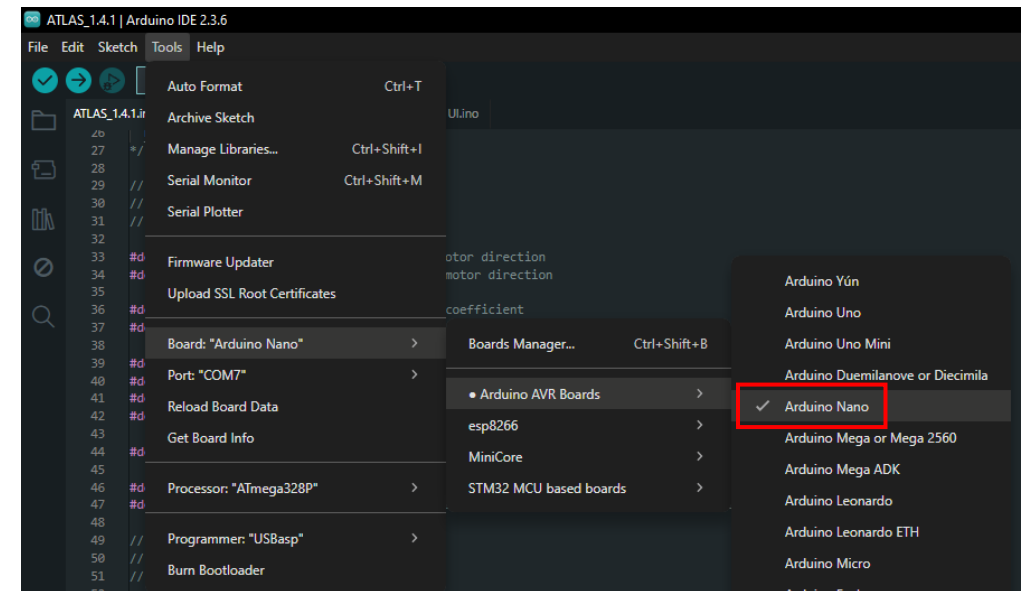
1. Have the Arduino IDE installed.
2. Have the WinUSB driver (v6.1.7600.16385) installed.
3. Have the USBasp programmer.
4. Open the ATLAS_1.4.1.ino file.

Robot flashing – IDE configuration

The ATLAS_1.4.1.ino file also includes several additional files with the extension “.ino,” which function as modules that separate and organize the robot's different functions. In this scheme, ATLAS_1.4.1.ino acts as the main file (equivalent to a “main.c”), and is where the user-configurable parameters are located, as well as the starting point of the program.

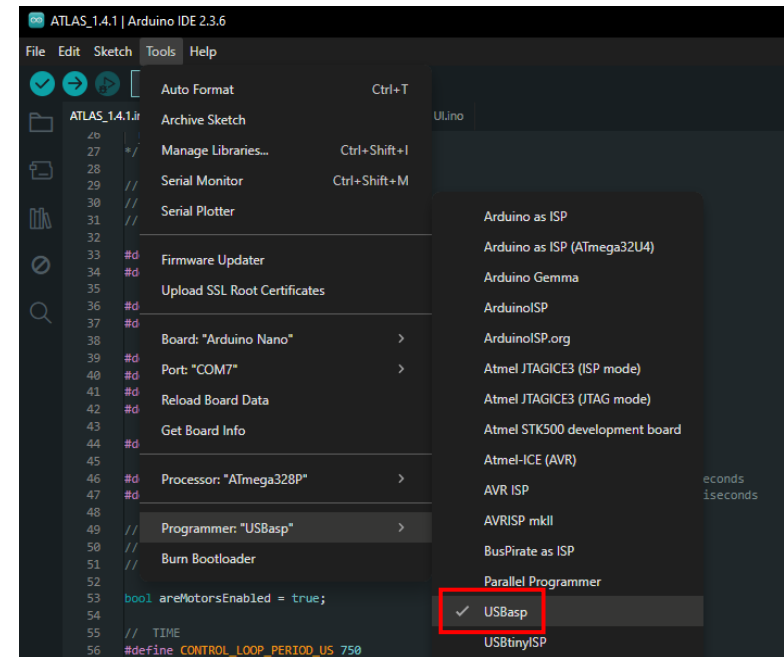
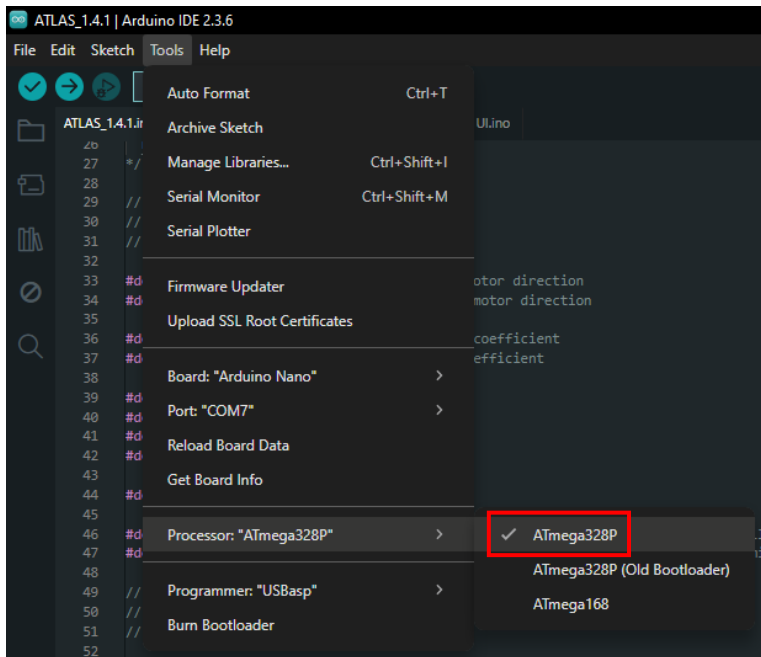
Once the Arduino IDE is open, a series of preliminary configurations must be made in order to flash the microcontroller using the USBasp programmer:

1. Select the corresponding board from the menu: Tools > Board > Arduino AVR Boards > Arduino Nano



Robot flashing – IDE configuration

2. Next, select the microcontroller: Tools > Processor > ATmega328P. You can use either the “ATmega328P” or “ATmega328P (Old Bootloader)” option, as both are compatible in this context.
3. Finally, select the programmer: Tools > Programmer > USBasp



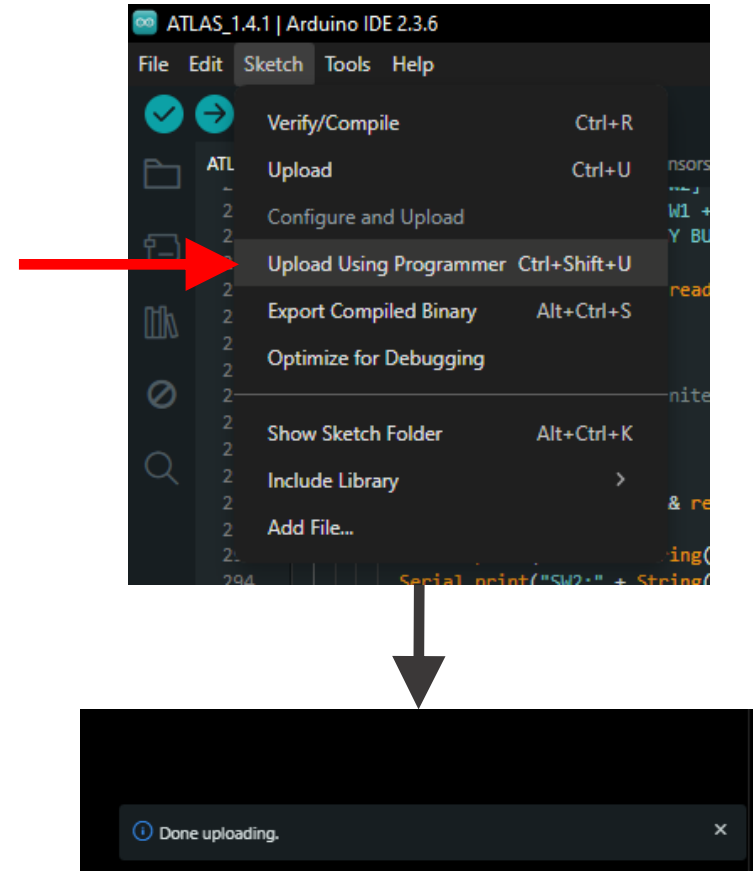
Robot flashing – USBasp as a programmer

Unlike the UART programming method used in conventional Arduino boards, in this case it is not necessary to select a serial communication port, as programming is done through the ICSP interface.

Once the Arduino IDE is correctly configured, proceed as follows:

1. Connect the USBasp programmer to both the computer and the ATLAS robot (the connection order is not relevant), ensuring that the connector polarity is correct.
2. After verifying and saving any modifications made to the code, go to the menu: Sketch > Upload Using Programmer or press the key combination Ctrl + Shift + U.
3. The Arduino IDE will compile the sketch and automatically start the flashing process.

The programming process takes approximately 5 to 15 seconds, depending on the performance of the equipment. Upon successful completion, the IDE will display the message “Done uploading,” indicating that the flashing has been successful.



Debug mode – Access

As mentioned above, the ATLAS_1.4.1.ino firmware incorporates an alternate diagnostic mode (debugMode), which allows direct testing of all system peripherals for verification, diagnosis, and functional validation purposes.

Access to this mode is achieved through a conditional statement within the setup() function during the system boot process, which, when fulfilled, will call the debugMode() function.

When the robot is turned on or the RESET button is pressed, immediately after LED2 completes its startup flashing animation and just before the calibration process is enabled, the firmware evaluates an input condition. If during that interval it detects that either the SW1 or SW2 button is pressed, the system will execute a confirmation animation and then enter diagnostic mode.

The recommended technique for reliably accessing this mode is to hold down either button (SW1 or SW2) while LED2 is flashing during program startup, until the corresponding confirmation animation is observed.

Initial animation

Condition

Calling

```
102
103 void setup() {
104     cli(); // disable global interrupts
105
106     UIInit();
107     sensorsInit();
108     motorsInit();
109
110     sei(); // enable global interrupts
111
112     bootAnimation(); // boot display animation
113
114     // debug mode trigger
115     if (readButton_1() || readButton_2()) {
116         confirmAnimation(500, 1);
117         debugMode();
118     }
119
120     // =====
121     // I D L E   S T A T E
122     // =====
123
124     while (!readButton_1() && !readButton_2())
125     ;
126
127     invertSensorReads = readButton_2();
128
129     calibrateSensors();
130     delay(250);
131
132     runningStart();
133 }
```

Debug mode – Use

The diagnostic mode includes four preset tests designed to verify the correct operation of the motors, sensors, and user interface. Additionally, the user can implement custom tests according to their specific needs, ensuring that the microcontroller's memory limits are not exceeded.

Once in this mode, the system enters an interactive loop that allows the user to select the type of test to be run. All available options, as well as system actions and statuses, are displayed through the serial communication port, which functions as a console.

To use this functionality, a USB–TTL converter must first be connected between the robot and the computer. In this case, the connection order is important:

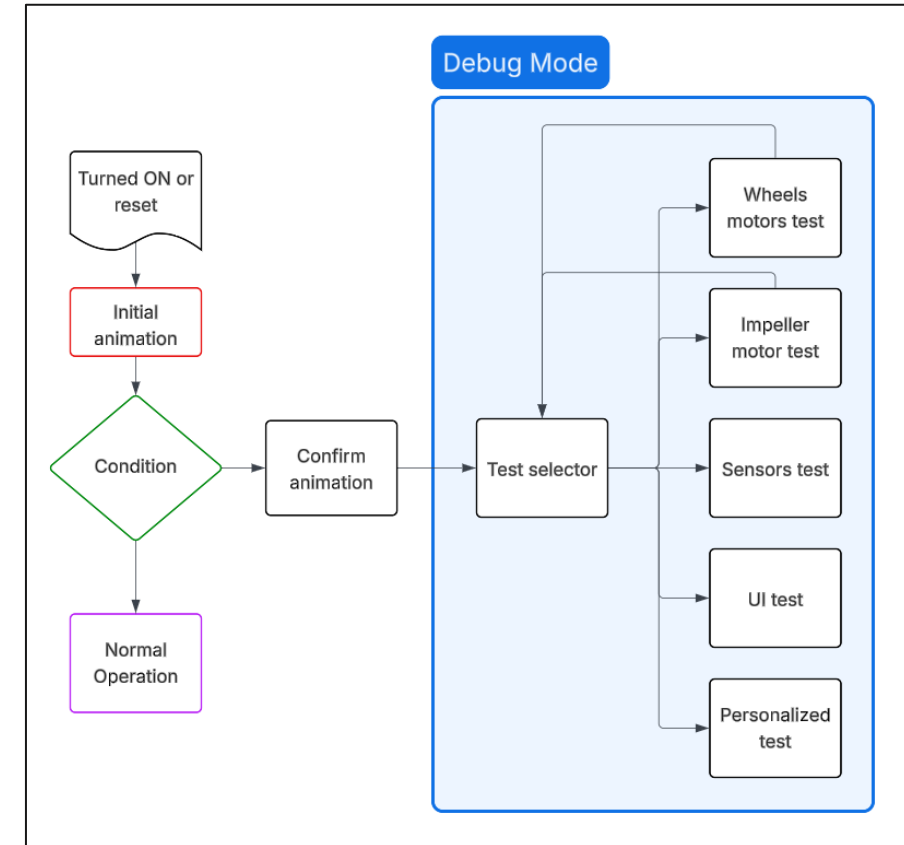
1. First connect the converter to the robot's UART port.
2. Then connect the converter to the computer and select the corresponding serial port in the Arduino IDE or serial monitor.

Debug mode – Flow diagram

Finally, you must open the serial monitor and verify that it has been connected correctly.

Below is a simplified flowchart describing the sequence of this mode.

During the execution of the tests, all instructions and warnings displayed on the console must be carefully followed in order to ensure the effectiveness of the diagnostics and avoid possible accidents or damage to the system.

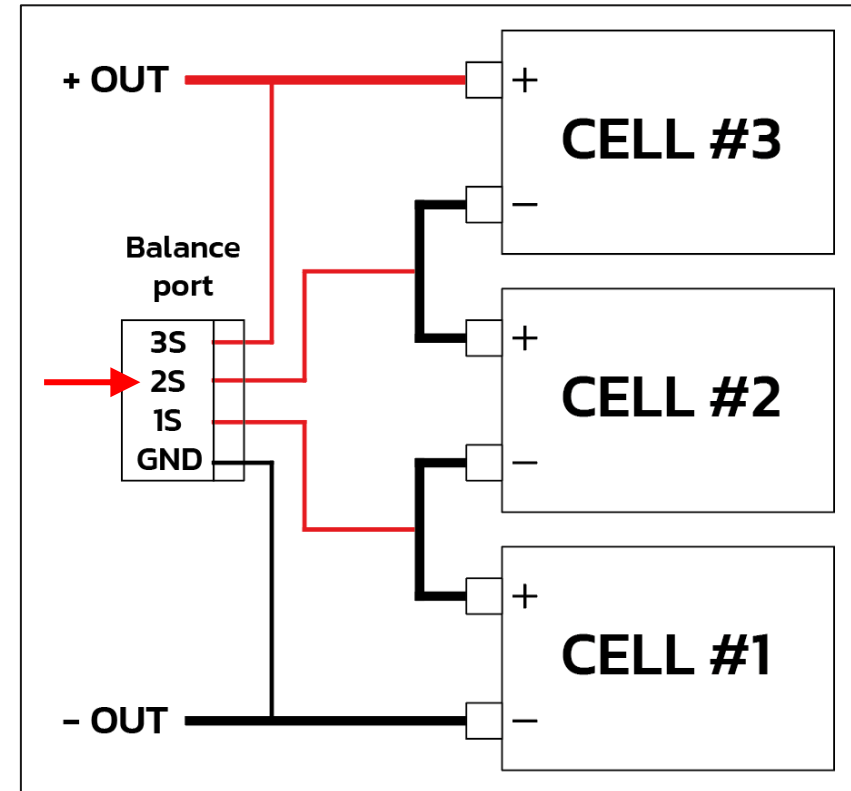


Robot operation – Turn ON

Before powering the robot, you must correctly fit the battery into its housing and connect the suction system cable to the battery balance connector, ensuring that the connection corresponds to a 2S (7.4 V) output. This step is critical, as an incorrect connection can cause the suction system to malfunction or, in extreme cases, destroy the impeller due to overvoltage.

If you are not completely sure which of the terminals corresponds to the 2S output, check the voltage using a meter before making the connection. Do not continue with the process until you have confirmed the correct value.

Once all connections have been secured and the suction system has been verified to be properly powered, you can proceed to connect the main battery to turn on the robot.



Robot operation – Calibration

Regardless of whether the user has reprogrammed the robot or is using the default firmware, the operating sequence of the ATLAS Rev. 1.4 model is always the same, unless that sequence has been explicitly modified in the code. This procedure must be performed each time the robot is to be put into operation.

It should be noted that both line calibration and speed selection are automatically reset when the robot is turned off or restarted, so these steps must be repeated at each startup.

Once the robot has been turned on or restarted, to start the calibration process, place the robot on the track surface, ensuring that the sensor bar is in parallel contact with the ground. To select the line color, press the SW1 button if the line is white, or the SW2 button if the line is black.

Once the selection is made, the calibration process will begin and the robot will behave as follows:

- If a black line is selected, LED0 and LED1 will flash.
- If a white line is selected, only LED2 will flash.

During the calibration process, a single slow and continuous pass over the line must be made, ensuring that the sensors do not separate from the surface to avoid invalid readings. It is recommended to apply slight pressure to the sensor bar rods during this process. To complete the calibration, without lifting the sensor bar, press either button (SW1 or SW2), which will trigger a confirmation animation and switch to speed selection or start mode.

Robot operation – Speed selection

The system starts with a base speed value of 100 (BASE_PWM_MOTORS). To modify this value:

- Press SW2 to increase the speed.
- Press SW1 to decrease the speed.

The speed can be adjusted within a range between 60 (MIN_VELOCITY_DUTY) and 250 (MAX_VELOCITY_DUTY), with increments or decrements of 10 units (VELOCITY_DUTY_INCREMENT) for each press.

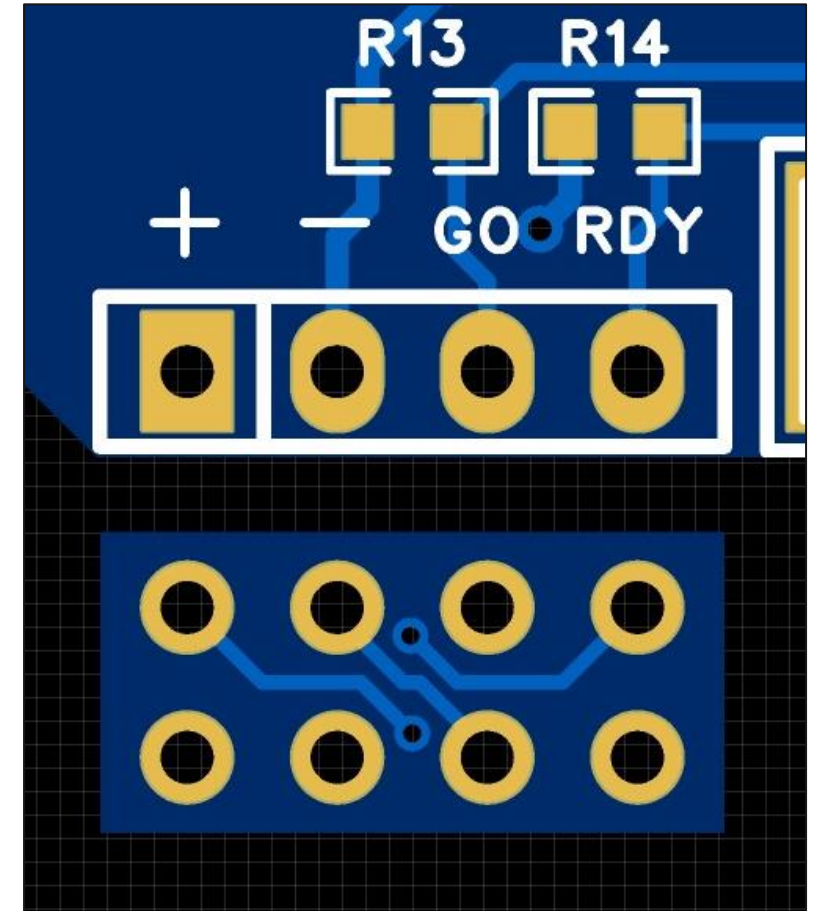
The suction system motor operates at a fixed value of 200 (PWM_IMPELLER), which cannot be modified from the robot interface. All these constants can be adjusted directly in the code, according to the user's needs.

Robot operation – Start Run

Once the desired speed has been selected, enabling the READY signal on the start module will initiate a PWM ramp for the suction system lasting 1 second (START_IMPELLER_RAMP_TIME_MS). When the maximum suction value (PWM_IMPELLER) is reached, LED2 will light up as visual feedback. At any time prior to the start of the race, the suction system can be deactivated by disabling the READY signal of the start module.

To start the race, the GO signal on the start module must be activated. Once the robot is in operation, simply deactivate this signal to stop it. When you do this, the robot will enter a stop state, which will be indicated by the signaling system flashing intermittently.

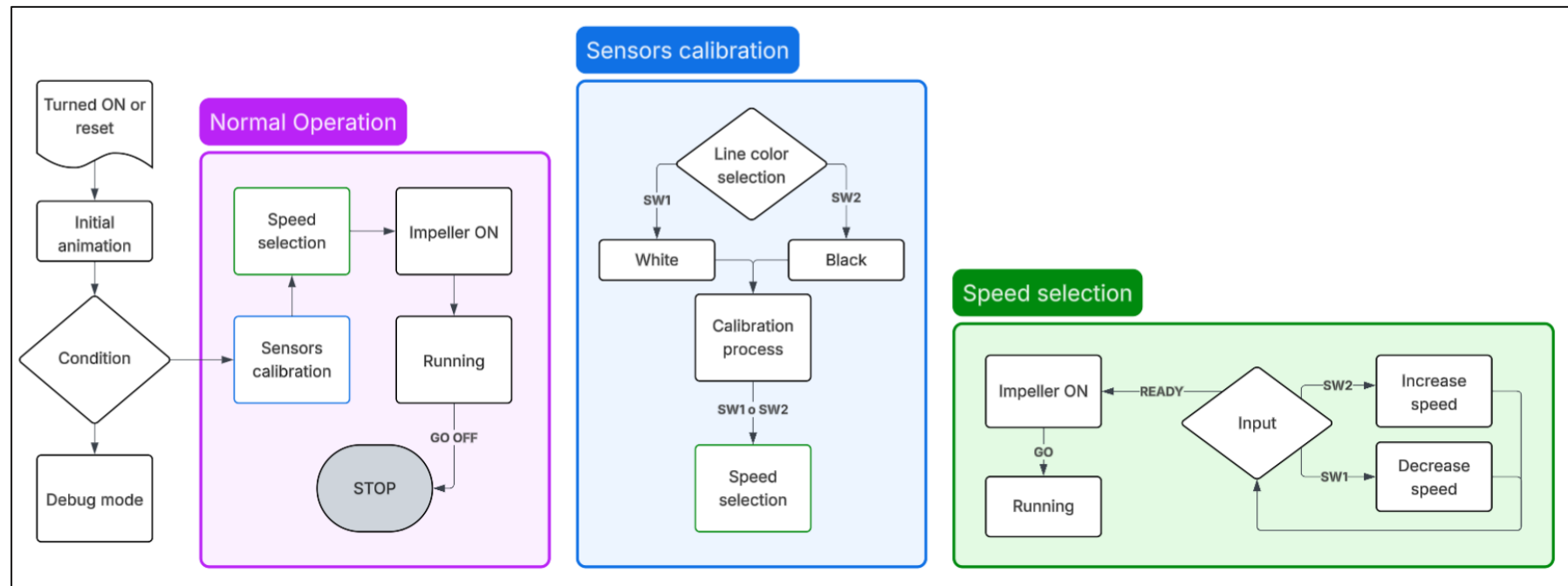
Ensure that the start module is connected correctly. The names of the corresponding pins can be found on the robot terminal. If your start module model has reverse polarity, you can use the adapter that comes panelized next to the USB-TTL converter connector.



Robot operation – Flow diagram

Below is a simplified flowchart describing the general sequence of operation of the ATLAS_1.4.ino firmware.

As mentioned previously, it is recommended to analyze and understand the sequence implemented in the source code to obtain greater feedback on each action and selection made during operation.



Warnings and recommendations – 1

As mentioned above, the ATLAS series has been designed as a technical tool for research, development, and experimentation. For this reason, it is essential to observe the following warnings and follow the recommendations described below in order to ensure proper operation, prevent possible accidents, and extend the robot's service life.

- Speed Parameters

A maximum value of 160 is recommended for speed selection. However, if a suboptimal response is detected during line tracking, a lower speed should be selected. The robot must follow the line in a stable manner and remain within it at all times, provided that the geometry of the track and the layout of the robot allow it.

It should be noted that the design characteristics of the ATLAS series allow it to reach extreme speeds. Failure to comply with this recommendation may result in loss of control, run-off, or catastrophic damage in the event of loss of control during operation.

- Suction Parameter

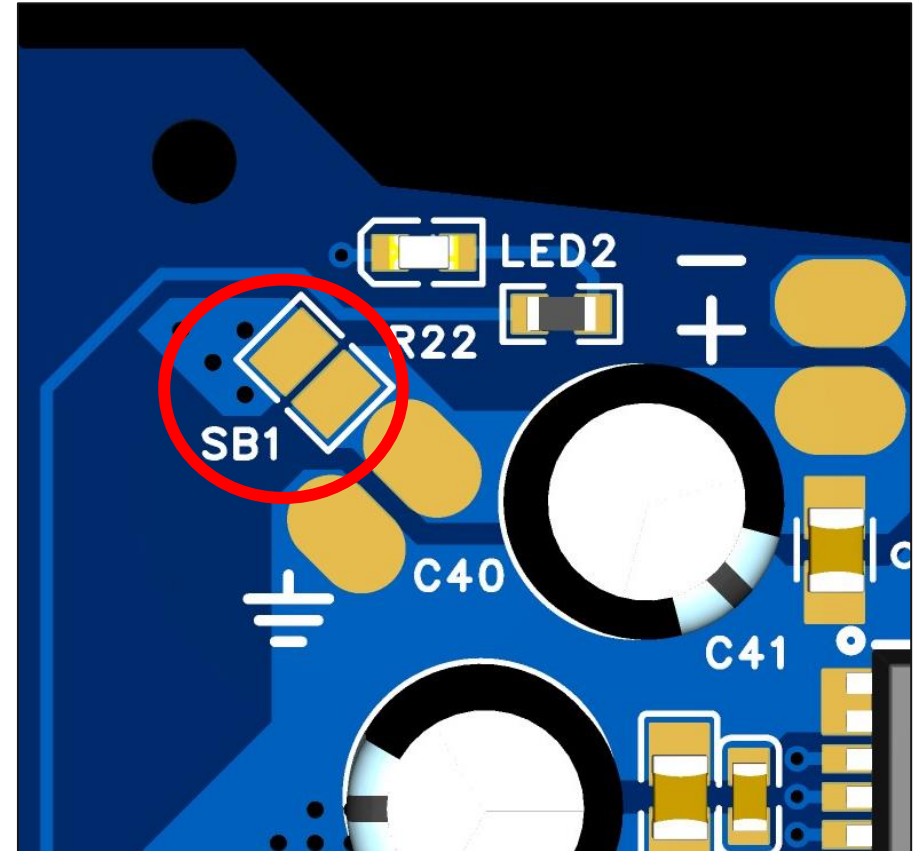
For the suction parameter (PWM_IMPELLER), a maximum value of 215 is recommended, which represents the optimal suction level considering the robot's speed range. Increasing this value can cause excessive current consumption, significantly affecting the system's autonomy, reducing the service life of the suction motor, or even deteriorating line tracking performance during operation.

Warnings and recommendations – 2

- Speed use and battery selection

It is recommended to operate the robot using the lowest possible speed value, according to the context of use, in order to extend the useful life of the system and reduce the risk of accidents. Likewise, when it is not strictly necessary to operate at high speeds, it is advisable to use a 2S battery, which is sufficient in most test scenarios.

When using a 2S battery, it is possible to join the SB1 connections and remove the cable corresponding to the suction motor connector. This eliminates the need to make this connection each time the robot is used, simplifying its operation.



Warnings and recommendations – 3

- Pre-session check

Before each test session, it is recommended to check that all system peripherals are working properly, or at least the critical components: motors, sensors, and suction system. If there is any suspicion or indication of abnormal operation, the robot should not be started until the problem has been corrected or its correct performance confirmed. Ignoring this recommendation may result in accidents or severe damage during operation.

- Suction skirt

Avoid touching or manipulating the suction skirt as much as possible, as this may permanently alter its geometry, causing inconsistent or insufficient suction. If it becomes dirty, it is recommended to remove dust using compressed air. If the dirt persists, a soft cloth slightly moistened with isopropyl alcohol can be used to clean the surface carefully.

Before each start-up, visually check that the suction surface is exerting normal and uniform force.

Warnings and recommendations – 4

- Operating voltage

For consistent results, it is recommended that the battery voltage does not fall below:

- 12.1 V in 3S configurations
- 8.1 V in 2S configurations

- Regulation and temperature

As specified in the technical characteristics section, the ATLAS Rev 1.4 model incorporates a 78M05 linear regulator, which can generate localized overheating. It is recommended not to keep the robot turned on for periods longer than 45 seconds (including the time while it is running). If high temperatures are detected, turn off the robot and allow it to cool down before continuing with further tests. Likewise, in 3S configurations, it is recommended not to exceed a test time of 8 seconds, otherwise the motors could be permanently damaged.

Warnings and recommendations – 5

- Wheel installation and maintenance

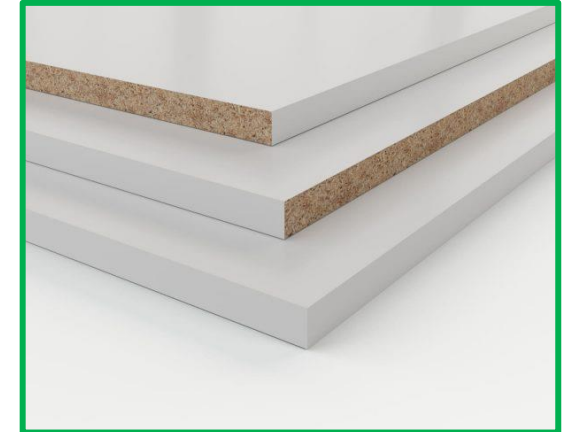
When installing wheels, ensure that the outer nut is not overtightened and verify that the wheel has approximately 0.5 mm of play along the axis.

To clean the wheels, use only masking tape to remove dust particles. Avoid touching the rubber with your hands, as even small amounts of body oil can negatively affect traction. Isopropyl alcohol can be used periodically to remove any residue that reduces grip.

- Track surface

The test surface should be flat, white or black, completely clean, and free of unevenness or pronounced joints. Recommended surfaces include melamine-coated boards, smooth-finished varnished surfaces, or adhesive vinyl films.

The use of materials such as vinyl canvas or banners does not guarantee adequate performance, and it is the user's responsibility to test on these types of surfaces.



Warnings and recommendations – 6

- Track operating conditions

The ATLAS series has been designed exclusively to operate on completely flat surfaces and in loops with diameters greater than 1.5 m, provided they are correctly constructed. Operation on ramps with angles greater than 10° or on steep slopes may damage the sensor bar or severely affect suction performance. Use of the robot under these conditions is the responsibility of the user, who must make the necessary modifications to protect the integrity of the robot.

- Wheel replacement

The wheels should only be replaced when they show significant wear or when the level of traction is no longer adequate. To remove them, push the rim from the outside and gradually remove the rubber, avoiding touching the gear. When installing a new rubber, it is recommended to apply isopropyl alcohol to the inside of the wheel to facilitate its placement.

- Motor overheating

Excessive or prolonged use can cause the motors to overheat, which could soften the PLA material of the mount, loosening the motor and allowing it to rotate on its own axis. If this occurs, it is recommended to apply an adequate amount of instant adhesive (cyanoacrylate), such as LOCTITE 401 or equivalent, to the joint between the motor and the mount.

Warnings and recommendations – 7

- Repairs and soldering

For repairs or component replacement, we recommend using MECHANIC Nano Flux Paste M35 and 60/40 solder with a melting point of 183 °C, using a soldering iron temperature appropriate for the component being replaced. Afterwards, clean the area with isopropyl alcohol to remove flux residue or contaminants.

- Environmental conditions

Avoid storing or leaving the robot in areas with direct or partial sun exposure, humid environments, or temperatures above 50 °C, as these conditions can cause structural deformation, especially in parts made of PLA.

- Test logging

As far as possible, it is recommended to record all tests performed, as this information is very useful for fault diagnosis, inconsistency analysis, and robot performance optimization.

- Support

If you require additional information or have any questions related to the ATLAS series, please do not hesitate to contact Exotic Team MX for support and additional guidance.

